

1) Lesson 1

1.1) Inequalities

General Procedure:

1. Determine the domain D .
2. Bring all terms to one side: $f(x) \geq 0$.
3. Factor into products or fractions: $\frac{A(x)}{B(x)} \geq 0$.
 1. Find the roots of each factor.
 2. Construct a sign chart.
4. Intersect the resulting intervals with the domain D .

There can happen 3 main cases: fractions, square roots or absolute values

Ex c) Fraction

Solve the following inequality:

$$\frac{x-1}{x-2} > \frac{2x-3}{x-3}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{2, 3\}$$

Simplify:

$$\frac{x-1}{x-2} - \frac{2x-3}{x-3} > 0 \implies \frac{(x-1)(x-3) - (2x-3)(x-2)}{(x-2)(x-3)} > 0$$

This is of form $A(x)/B(x)$ so the study of the numerator and denominator needs to be done independently

Numerator:

Find zeroes:

$$\begin{aligned}(x-1)(x-3) - (2x-3)(x-2) &= (x^2 - 4x + 3) - (2x^2 - 7x + 6) \\ &= -x^2 + 3x - 3\end{aligned}$$

Recall the quadratic formula:

Let $a, b, c \in \mathbb{R}$, then $ax^2 + bx + c = 0$ has the following solution:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

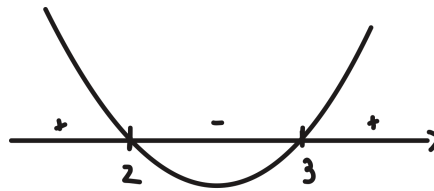
Where the discriminant $\Delta = b^2 - 4ac$ determines the types of solution:

- $\Delta > 0$: two real (distinct) solutions
- $\Delta = 0$: two real (identical) solutions
- $\Delta < 0$: no real solution

In this case since $\Delta = 9 - 12 = -3 < 0$ we don't have any solutions, hence since $a < 0$ the whole numerator is negative.

Denominator:

Here the function already shows the zeroes, so we can plot the graph to see the regions:



Curve

So we have positive denominator for $x < 2 \cup x > 3$.

By noticing the interval where they have the same sign:

x	$(-\infty, 2)$	$(2, 3)$	$(3, +\infty)$
Numerator	-	-	-
Denominator	+	-	+
Fraction	-	+	-

Finally we have:

$$\boxed{2 < x < 3}$$

Ex d) Absolute value

Solve the following inequality:

$$\frac{|x|}{x-1} > \frac{x+1}{2x-1}$$

Domain:

$$D = x \in \mathbb{R} \setminus \left\{ \frac{1}{2}, 1 \right\}$$

Now recall the abs value definition:

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

In this case $f(x) = x$ so we have:

$$\begin{cases} \frac{x}{x-1} > \frac{x+1}{2x-1} & x \geq 0 \\ \frac{-x}{x-1} > \frac{x+1}{2x-1} & x < 0 \end{cases}$$

Remember:

- The solution to each part of the system must be put in intersection with the condition
- The solutions of the parts of a system are in union with each other

Start with case 1:

Rewrite so that it is a single fraction:

$$\frac{x}{x-1} - \frac{x+1}{2x-1} > 0 \implies \frac{x(2x-1) - (x+1)(x-1)}{(x-1)(2x-1)} > 0$$

- Numerator:

$$x(2x - 1) - (x^2 - 1) = 2x^2 - x - x^2 + 1 = x^2 - x + 1$$

Find the zeroes of the numerator:

$$x^2 - x + 1 = 0 \rightarrow \Delta = -3$$

Since $\Delta < 0$ and $a = 1 > 0$, the numerator is strictly positive.

- Denominator:

$$(x - 1)(2x - 1) > 0 \iff x < \frac{1}{2} \cup x > 1$$

Sign analysis for Case 1:

x	$(-\infty, \frac{1}{2})$	$(\frac{1}{2}, 1)$	$(1, +\infty)$
Numerator	+	+	+
Denominator	+	-	+
Fraction	+	-	+

So the fraction is > 0 on $(-\infty, \frac{1}{2}) \cup (1, +\infty)$.

Intersecting with the case condition $x \geq 0$:

$$S_1 = \left[0, \frac{1}{2}\right) \cup (1, +\infty)$$

Case 2:

Same procedure as case 1:

$$\frac{-x}{x-1} - \frac{x+1}{2x-1} > 0 \implies \frac{-x(2x-1) - (x+1)(x-1)}{(x-1)(2x-1)} > 0$$

- Numerator:

$$-2x^2 + x - (x^2 - 1) = -3x^2 + x + 1$$

Find the zeroes of the numerator:

$$x_{1,2} = \frac{-1 \pm \sqrt{13}}{-6} = \frac{1 \mp \sqrt{13}}{6}$$

Since $a = -3 < 0$, the parabola opens downward:

$$N > 0 \iff x_1 = \frac{1 - \sqrt{13}}{6} < x < \frac{1 + \sqrt{13}}{6} = x_2$$

Denominator:

The denominator roots are $\frac{1}{2} = 0.5$ and 1.

Sign analysis for the factors:

x	$(-\infty, x_1)$	$(x_1, 1/2)$	$(1/2, x_2)$	$(x_2, 1)$	$(1, +\infty)$
Numerator	-	+	+	-	-
Denominator	+	+	-	-	+
Fraction	-	+	-	+	-

The fraction is > 0 on $(x_1, \frac{1}{2}) \cup (x_2, 1)$.

Intersecting with the case condition $x < 0$:

$$S_2 = \left(\frac{1 - \sqrt{13}}{6}, 0 \right)$$

Taking the union of both cases $S = S_1 \cup S_2$:

$$S = \left(\frac{1 - \sqrt{13}}{6}, 0 \right) \cup \left[0, \frac{1}{2} \right) \cup (1, +\infty)$$

Notice that 0 is included in the solution set, so the first two intervals join together:

$$S = \left(\frac{1 - \sqrt{13}}{6}, \frac{1}{2} \right) \cup (1, +\infty)$$

Ex f) Square Root

Solve the following inequality:

$$\sqrt{x^2 - 6x} > x + 2$$

First notice that the domain is

$$x^2 - 6x \geq 0 \rightarrow D = (-\infty, 0] \cup [6, +\infty)$$

First notice that when $x + 2 < 0$ the solution inequality holds (in the domain). So one part of the solution is:

$$S_1 = D \cap \{x < -2\} = \{x < -2\}$$

Now for $x + 2 \geq 0$ of both sides can be squared:

$$x^2 - 6x > x^2 + 4x + 4 \rightarrow 0 > 10x + 4 \rightarrow x < -\frac{2}{5}$$

This clearly for the values bigger than -2 in the domain, that is

$$S_2 = D \cap \left\{ \{x \geq -2\} \cap \left\{ x < -\frac{2}{5} \right\} \right\} = D \cap \left\{ -2 \leq x < -\frac{2}{5} \right\} = \left\{ -2 \leq x < -\frac{2}{5} \right\}$$

So the result is:

$$S = S_1 \cup S_2 = \{x < -2\} \cup \left\{ -2 \leq x < -\frac{2}{5} \right\} = x < -\frac{2}{5}$$

Ex i) Mixed Together

Solve the following inequality:

$$\sqrt{|x^2 - 4|} - x \geq 0$$

Domain: $|x^2 - 4| \geq 0 \rightarrow D = \mathbb{R}$

Now recall the abs value definition:

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

In this case $f(x) = x^2 - 4 = (x + 2)(x - 2)$ so $f(x) \geq 0 \rightarrow x \leq -2 \cup x \geq 2$

Now we can rewrite the inequality as a system of two inequalities:

$$\sqrt{|x^2 - 4|} - x \geq 0 \rightarrow \begin{cases} \sqrt{x^2 - 4} - x \geq 0 & x \leq -2 \cup x \geq 2 \\ \sqrt{4 - x^2} - x \geq 0 & -2 < x < 2 \end{cases}$$

Each system can be solved independently from the other.

Call the conditions of the cases D_1 and D_2 respectively.

Solve first case:

Recall that when RHS (x) is negative there is always a solution.

$$\sqrt{x^2 - 4} \geq x$$

Therefore one part of the solution includes $x < 0$

Now study in the interval $x \geq 0$: since we are sure that both sides are ≥ 0 we can square both sides:

$$x^2 - 4 \geq x^2 \rightarrow -4 \geq 0$$

Not possible, so the only solution to this part is when x is negative:

$$S_1 = x \leq 0 \cap D_1 = x \leq -2$$

Solve the second equation:

As before a solution always exists for $x < 0$. Now consider only the positive values ($x \geq 0$) in order to be able to square both sides:

$$\sqrt{4 - x^2} \geq x \rightarrow 4 - x^2 \geq x^2 \rightarrow 4 - 2x^2 \geq 0 \rightarrow -\sqrt{2} \leq x \leq \sqrt{2}$$

So we have $S_2 = D_2 \cap \{x < 0 \cup -\sqrt{2} \leq x \leq \sqrt{2}\} = D_2 \cap x \leq \sqrt{2} = -2 < x \leq \sqrt{2}$

The result is therefore the union of both cases which yields:

$$\boxed{S = x \leq -2 \cup -2 < x \leq \sqrt{2} \rightarrow S = x \leq \sqrt{2}}$$

1.2) Subsets of $\mathbb{R}$

Ex a)

Describe the following set:

$$A = S_1 \cap S_2 = \{x \in \mathbb{R} : x^2 + 4x + 13 < 0\} \cap \{x \in \mathbb{R} : 3x^2 + 5 > 0\}$$

Subset 1:

$$x^2 + 4x + 13 < 0 \rightarrow \Delta = -36 \implies S_1 = \emptyset$$

any intersection with an empty set is an empty set:

$$\boxed{A = \emptyset \cap S_2 = \emptyset}$$

In general recall:

$$\emptyset \cap S = \emptyset \quad \emptyset \cup S = S$$

Ex c)

Describe the following set:

$$C = \left\{ x \in \mathbb{R} : \frac{x^2 - 5x + 4}{x^2 - 9} < 0 \right\} \cup \left\{ x \in \mathbb{R} : x + \sqrt{7x + 1} = 17 \right\}$$

Subset 1:

First notice that the domain is $x \neq \pm 3$

Numerator:

$$x^2 - 5x + 4 = 0 \rightarrow x_{1,2} = 1, 4 \rightarrow N > 0 \iff x < 1 \cup x > 4$$

Denominator:

$$x^2 - 9 = (x + 3)(x - 3) = 0 \rightarrow D > 0 \iff x < -3 \cup x > 3$$

x	$(-\infty, -3)$	$(-3, 1)$	$(1, 3)$	$(3, 4)$	$(4, +\infty)$
Numerator	+	+	-	-	+
Denominator	+	-	-	+	+
Fraction	+	-	+	-	+

so $S_1 = (-3, 1) \cup (3, 4)$

Subset 2:

This is just an equality:

$$\sqrt{7x + 1} = 17 - x$$

The domain is $7x + 1 \geq 0 \rightarrow D = \left\{ x \geq -\frac{1}{7} \right\}$

Moreover as before, both sides must have the same sign when squaring and since the sqrt is always ≥ 0 we must have that also $17 - x \geq 0 \rightarrow x \leq 17$ so we can further reduce the domain to

$$D = \left[-\frac{1}{7}, 17 \right]$$

Square both sides:

$$7x + 1 = (17 - x)^2 \rightarrow 7x + 1 = x^2 - 34x + 17^2 \rightarrow x^2 - 41x + 288 = 0$$

$$x_{1,2} = 9, 32$$

But only 9 is acceptable as $32 \notin D$

Finally:

$$C = (-3, 1) \cup (3, 4) \cup \{9\}$$

1.3) Max/Min, Sup/Inf

Recall the theory

Definition (Upper/Lower Bound).

$M \in \mathbb{Q}$ is called **upper bound** for A if $M \geq a, \forall a \in A$.

Respectively it is called **lower bound** if $M \leq a, \forall a \in A$

Notice M does not need to be in A

Example: consider set $A = [0, 1)$ any $m \leq 0$ is a lower bound and any $M \geq 1$ is upper bound

Definition (Maximum/Minimum of A).

If $\overline{M}$ is an upper bound for A and $\overline{M} \in A$, then $\overline{M}$ is the *minimum* possible *upper* bound and is the **maximum** (of A).

If $\overline{m}$ is a lower bound for A and $\overline{m} \in A$, then $\overline{m}$ is the *maximum* possible *lower* bound and the **minimum** (of A).

In this case $\overline{M} \in A$ by definition.

From the previous example: $\overline{m} = \max \{m \leq 0\} = 0 \in A$ is the correct minimum of A , however $\overline{M} = \min \{M \geq 1\} = 1 \notin A$ means that we don't have a maximum

Definition (Supremum/Infimum).

The **supremum** of A $\sup A$ is the minimum of the upper bounds of A .

The **infimum** of A $\inf A$ is the maximum of the lower bounds of A .

Finish the example: we know $\overline{m} = 0 \in A$ so this is both minimum and infimum of A . We know also $\overline{M} = 1 \notin A$, therefore $\sup A = 1$ but it is not the maximum.

Recall that if a set is unbounded, then

$$\sup A = +\infty \quad \text{if } A \text{ is unbounded from above}$$

$$\inf A = -\infty \quad \text{if } A \text{ is unbounded from below}$$

Ex a)

Tell whether this subset is bounded and the sup/inf and if it allow for maximum/minimum.

$$A = \left\{ x \in \mathbb{R} : x = n \text{ or } x = \frac{1}{n^2}, n \in \mathbb{N} \setminus \{0\} \right\}$$

It is immediately clear that A is the union of two sets, mainly

$$A = S_1 \cup S_2 \text{ where } \begin{aligned} S_1 &= \{n \in \mathbb{N} \setminus \{0\}\} = \{1, 2, 3, 4, \dots\} \\ S_2 &= \left\{ \frac{1}{n^2} \mid n \in \mathbb{N} \setminus \{0\} \right\} = \left\{ 1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \dots \right\} \end{aligned}$$

Find the bounds on both sets individually (not required but the result is clearer).

Study the first subset S_1 : notice that it is just the natural numbers with zero excluded, so its properties are already known.

- No upper bound, so also no max
 - Supremum: since unbounded; $\sup S_1 = +\infty$
- Lower bound: $m_1 \leq 1$
 - Minimum: $\overline{m}_1 = \max m_1 \leq 1 = 1 \in S$

Study the second subset S_2 : Intuitively we know that the function describing the set is monotonically decreasing: we can express it by saying:

Call x_n the value of x for any chosen n , then $x_{n+1} < x_n$.

This is easily shown as the following inequality is true for $n \in \mathbb{N} \setminus \{0\}$ (check at home):

$$\frac{1}{(n+1)^2} < \frac{1}{n^2}$$

Therefore for every number $x_n > 0$ we can find a smaller number such that $0 < x_{n+1} < x_n$

Is 0 the maximum lower bound?

By contradiction suppose there exists a lower bound $\delta > 0$ then if 0 is the maximum lower bound we have

$$\exists n \in \mathbb{N} \setminus \{0\} \quad \text{such that} \quad \frac{1}{n^2} < \delta$$

Now compute

$$\frac{1}{n^2} < \delta \rightarrow n^2 > \frac{1}{\delta} \rightarrow n > \frac{1}{\sqrt{\delta}}$$

Because $\delta > 0$, the number $\frac{1}{\sqrt{\delta}}$ is just a fixed positive real number and an appropriate n exists.

With this digression aside, we also know that $0 \notin S_2$.

As before analyze:

- Lower bound: $m_2 \leq 0$
 - No minimum: $\overline{m}_2 = \max m_2 \leq 0 = 0 \notin S_2$
 - Infimum is $\inf S_2 = 0$
- Upper bound: $M_2 \geq 1$
 - $\overline{M}_2 = \min M_2 \geq 1 = 1 \in S_2$

Putting all together:

Now consider the union, so:

- No upper bound since S_1 is unbounded
 - therefore $\sup A = +\infty$
- Lower bound: The infimum is the **maximum** of all lower bounds: $\inf A = \min(\inf S_1, \inf S_2) = \min(1, 0) = 0$. Since $0 \notin A$, $\min A$ does not exist.

2) Symbols

- $\implies$: implies, sufficient condition ($P \implies Q$ means P is sufficient for Q , Q is necessary for P).
- $\impliedby$: implied by, necessary condition
- $\iff$: necessary + sufficient, if and only if "iff"
- $\emptyset$: empty set
- $\mathbb{N}$: natural numbers; $\{0, 1, 2, \dots\}$
- $\mathbb{Z}$: integer numbers; $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- $\mathbb{Q}$: rational numbers; $\{\frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0\}$
- $\mathbb{R}$: real numbers
- $\mathbb{C}$: complex numbers
- $\cup$: Union; or
- $\cap$: Intersection; and
- $\exists$: there exists